VIGIL: Validity-Interval Gating of Legal Precedent for Product–Litigation Risk Entailment

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Abstract

Retrieval-augmented generation (RAG) treats semantically similar documents as interchangeable evidence, but in law a lexically close precedent may already have been narrowed, superseded, or overruled at the date that matters. We introduce Product–Litigation Risk Entailment (PLRE), a directional task that asks whether an AI product's attributes entail exposure to a legal-risk theory under precedent valid at a given query date. Our system, VIGIL (Validity-Interval Gating in Litigation), extracts treatment edges (overrules, supersedes, narrows) from later opinions, turns them into validity intervals, and validates them against gold citation spans (negative-treatment micro-F1 0.80) and a commercial citator (0.81). A simple retrieval-time gate applies these intervals before generation, and a goal-conditioned agentic loop stops when evidence coverage, not legal correctness, is satisfied. On VIGIL-Dataset (1,247 U.S. AI-litigation cases, 312 product-mapping cases), validity gating raises held-out PLRE F1 from 0.72 to 0.78 and cuts the stale-citation rate from 12.7% to 4.1% relative to the same pipeline without the gate (point estimates, not significance-tested). We release the dataset, label provenance, guidelines, and evaluation scripts. VIGIL is decision support for qualified reviewers, not legal advice.

1 Introduction

Generative-AI products face growing litigation over training data, output reproduction, model behavior, and downstream misuse (Henderson et al., 2023; Levendowski, 2018; Grimmelmann, 2016). High-profile suits such as *Getty Images v. Stability AI* (Getty Images (US), Inc., 2023), *New York Times v. Microsoft* (The New York Times Company, 2023), and *Kadrey v. Meta* (Kadrey et al., 2023) illustrate the range of theories from DMCA §1202 information removal to direct infringement and vicarious liability for outputs. For the engineering organizations that build these systems, the operational question is not merely whether a lawsuit exists, but whether a theory of liability maps onto a specific product attribute, whether the supporting authority remained valid at the relevant date, and which mitigation should be reviewed by counsel. Legal-operations teams track dockets and news, yet the bridge from a litigation event to a product-specific engineering action is usually manual. Keyword alerts surface noisy filings, docket trackers rarely map cases to product features, and conventional retrieval-augmented generation (RAG) (Lewis et al., 2020) treats semantically similar legal documents as interchangeable even when later decisions have narrowed or superseded them.

We formalize this need as Product–Litigation Risk Entailment (PLRE): deciding whether product attributes entail exposure to a legal-risk theory under time-valid precedent. PLRE differs from standard legal textual entailment (Rabelo et al., 2022) in three respects: the premise is an engineering description rather than legal text, the relation is directional (product attribute $\rightarrow$ risk theory), and the support set is conditioned on which authority was operative at the query date. Unlike commercial citators such as Shepard's (LexisNexis, 2024) and KeyCite (Thomson Reuters, 2024), we construct a public-corpus, automatically extracted, time-indexed treatment layer and connect it directly to product-facing entailment.

We present VIGIL (Validity-Interval Gating in Litigation), a litigation-risk analysis system built around this task; its document-sensing front end is infrastructure that we do not evaluate. Its core component is a supersession-aware treatment layer that infers when each authority ceases to be operative from treatment edges in later opinions. A simple retrieval-time gate applies these intervals before generation, rather than flagging stale citations after an LLM has already produced a report. Around this layer, a goal-conditioned agentic loop

exposes unmet evidence requirements and stops when a coverage threshold is met. Although the treatment-layer construction is in principle applicable to any time-stamped corpus in which later documents can supersede earlier ones, we evaluate it only for U.S. case law within VIGIL; external-stack transfer remains a pilot and is not claimed.

Our contributions are:

- a supersession-aware treatment layer that extracts treatment edges from public opinions, validates them against gold spans and a commercial citator, and propagates them into validity intervals that gate retrieval at query time (§3.4);
- PLRE, an initial product-conditioned legal-risk entailment benchmark with non-VIGIL baselines and ablations that separate temporal validity from flat similarity (§3.5, §5);
- VIGIL-Dataset with per-record provenance and a human-adjudicated subset (§4), plus a public SCOTUS overruling testbed built from the transfer pilot (Appendix H).

The goal-conditioned agentic loop that hosts the gate is a system component, not a separate contribution: without the gate it does not improve PLRE, and its conformal omission-risk bound holds only under exchangeability and is violated under temporal shift (§2.1, §5.4).

2 Problem Formulation

Let $\mathcal{D} = \{d_1, \ldots, d_n\}$ be public litigation documents with metadata and text, and let P be a product specification describing architecture, data provenance, modality, and deployment context. VIGIL returns a structured risk report R^* and mitigation candidates M^* that must satisfy an auditable goal predicate $\mathcal{G}$ (§2.1). Report quality is not optimized directly; we evaluate the evidence-coverage loss L_τ (§5.4) and report severity-label agreement and blinded report ratings in Appendix A. The core difficulty is a domain gap: product specifications employ engineering terms (data pipelines, API distribution), whereas litigation documents speak in causes of action, procedural posture, and precedent. PLRE (§3.5) therefore requires mapping engineering-language attributes onto legal-risk theories while conditioning the support set on authority that remains valid at the query date.

2.1 Goal Predicate and Stopping Rule

VIGIL treats report generation as goal-conditioned planning. The predicate is

$$\mathcal{G}(R) = \mathcal{G}_{\text{risk}}(R) \wedge \mathcal{G}_{\text{evidence}}(R) \wedge \mathcal{G}_{\text{guide}}(R), \quad (1)$$

where $\mathcal{G}_{\text{risk}}$ requires every risk item to carry a severity label, $\mathcal{G}_{\text{evidence}}$ requires at least k supporting documents above relevance threshold τ, and $\mathcal{G}_{\text{guide}}$ requires mitigation text for each High or Medium item. The controller selects among SEARCH, ANALYZE, SUMMARIZE, MAP, VERIFY, and REPORT; the deterministic (LLM-free) Goal Checker either terminates, emits a structured critique that closes residual gaps, or escalates unmet conditions.

Let $\Phi(R_t)$ count the unmet atomic sub-conditions of $\mathcal{G}$, and let $\mathcal{T}_c^H$ be the human-labeled High-risk theories of case c that are retrievable—i.e., supported by at least one authority in $\mathcal{D}$ that is valid at t_q. The coverage loss at evidence threshold τ is

$$L_\tau(c) = \mathbf{1}\left[\exists\, T \in \mathcal{T}_c^H : T \notin R^*\right]. \quad (2)$$

Observation 1 (Coverage-controlled stopping with escalation)**.** *Assume the set of risk items under consideration is fixed within a run (the Reporter may rewrite mitigation text but does not introduce new risk atoms that expand Φ). Under monotone evidence caching and attainable progress, $\Phi(R_t)$ is non-increasing, reaches zero in at most $\Phi(R_0)$ successful progress steps when all atoms are attainable, and otherwise returns the residual atoms at $T_{\max}$. If $\hat{\tau}$ is selected on exchangeable calibration cases by conformal risk control, then the next-case omission risk satisfies $\mathbb{E}[L_{\hat{\tau}}] \leq \alpha$.*

The first clause follows almost directly from the definitions (proof in Appendix F). The second clause is a standard application of conformal risk control (Angelopoulos et al., 2024) to L_τ, which is monotone in τ. It governs evidence coverage only, holds only for exchangeable cases—our temporal split violates it at $\alpha \in \{0.05, 0.10\}$ (§5)—and its tightness depends on the calibration-set size (150 cases).

3 Method: The VIGIL Framework

3.1 Overview

VIGIL comprises two layers. A sensing pipeline discovers AI-litigation filings in CourtListener/RECAP (Free Law Project, 2023) and legal news and converts them into case records; this

engineering infrastructure is documented in the supplementary material and is not evaluated here. The remainder of this section describes the evaluated NLP components: LegalRisk-RAG (§3.2), the goal-conditioned agentic loop (§3.3), the temporal precedent graph (§3.4), and the PLRE decision procedure (§3.5).

3.2 LegalRisk-RAG

The LegalRisk-RAG pipeline adapts standard RAG to legal text in three respects. **Chunking** segments documents at logical structural boundaries—numbered claims, "WHEREFORE" clauses, section headings—rather than fixed token windows; each chunk retains a metadata header (case identifier, filing date, court level, section type). **Embedding:** we compared a dense retriever (text-embedding-3-large), RECAP-tuned LegalBERT (Chalkidis et al., 2020), and a hybrid BM25 (Robertson and Zaragoza, 2009) + neural re-ranker on a 200-pair development set; the hybrid performed best and is the default. **Dynamic context expansion (precedent chaining):** when a retrieved chunk references a prior case, the pipeline appends that opinion as secondary context and queries the temporal precedent graph for subsequent treatment, so that the Analyzer receives both semantic neighbors and validity metadata.

3.3 Goal-Conditioned Agentic Loop

The four specialized agents and the central Goal Checker run until $\mathcal{G}$ holds or $T_{\max}$ is reached (hyperparameters in Appendix G). The Planner decomposes unmet sub-goals into ordered retrieval/analysis sub-tasks; the Retriever runs LegalRisk-RAG queries and reformulates via legal entity term expansion when $\mathcal{G}_{\text{evidence}}$ is unmet; the Analyzer scores each retrieved document d_i against P, clusters evidence, and maps clusters to a risk taxonomy (Training Data, Output Reproduction, API Misuse, Model Weights); the Goal Checker evaluates $\mathcal{G}(R_t)$ after each pass and returns machine-readable residuals; the Planner converts these critiques into expanded entity/date/theory queries; and the Reporter assembles the risk register, per-risk evidence with citations, and mitigation guidance.

3.4 Temporal Precedent Graph

VIGIL's central component is a treatment layer that determines when each authority ceases to be operative; a simple retrieval-time gate then applies it. The layer is realized as a temporal precedent graph $\mathcal{P}_t = (V, E_t)$. Each node $v \in V$ denotes a case, order, or doctrinal rule and stores jurisdiction, court level, issue tags, filing and decision dates, and a validity interval $[s_v, e_v)$, where e_v remains open unless the node is later vacated, overruled, or superseded; material narrowing instead lowers its treatment weight. Directed edges encode treatment relations: CITES, DISTINGUISHES, NARROWS, SUPERSEDES, and OVERRULES. Retrieval for a query at date t_q first obtains semantically relevant candidates and then applies validity gating:

$$\text{score}(v, q, t_q) = \text{sim}(v, q) \cdot \mathbf{1}[s_v \le t_q < e_v] \cdot w(\text{treat}(v, t_q)). \quad (3)$$

Here $w(\cdot)$ is a fixed monotone schedule: positively treated or still-open authorities receive weight 1.0; cautionary authorities (labeled DISTINGUISHES, or bearing weak negative treatment without interval closure) receive 0.75; material NARROWS receives 0.50; fully closed (overruled or superseded) authorities are already excluded by the indicator. The operator separates legal authority from topical similarity: a stale but lexically close case may still appear as context, yet it cannot support a risk decision without passing the validity check. Equation (3) is therefore a constrained retrieval operator over (q, t_q): a host retriever scores semantic relevance as usual, and PLRE receives only support whose legal status remains operative at t_q.

Why not a date filter? Once intervals are known, Eq. (3) is a metadata filter any vector store can apply. The difficulty is upstream and conceptual: e_v is set by later documents that overrule, supersede, or narrow v, so it is not recoverable from v's own filing date. A pure recency or document-date filter therefore cannot implement the same operator; the empirical question is whether treatment edges can be extracted reliably enough to use. Table 1 addresses that question: negative-treatment micro-F1 is 0.80 on 600 gold spans and 0.81 against a commercial citator on 400 sampled edges (Protocol P2; missed-action rate 20.4%; 3.8% of held-out PLRE pairs flip after citator correction; Appendix B–C).

Graph construction. Nodes are induced from CourtListener/RECAP with intervals initialized to $[s_v, \infty)$. Edges are extracted by a citation-span detector plus an LLM classifier (GPT-4o, $T = 0$); interval-closing edges must be anchored to a detected span, decision date, and source document,

Reference	P	R	F1
Gold spans ($n = 600$)	0.84	0.77	0.80
Citator, P2 ($n = 400$)	0.83	0.80	0.81

Table 1: Treatment-layer validation (neg.-treatment micro over OVERRULES/SUPERSEDES/material NARROWS; per-relation results in Appendix B).

and gold evaluation uses spans held out from prompts. A confirmed OVERRULES/SUPERSEDES edge at t sets $e_v \leftarrow t$; material NARROWS sets $w = 0.50$ from t. The graph has 1,812 nodes and 4,760 edges. Low-confidence negative-treatment edges are routed to human review rather than auto-closing intervals.

3.5 PLRE Decision Procedure

We cast product-litigation alignment as a risk-entailment task. Given a product spec P with attributes $\{p_1, \ldots, p_m\}$, a legal-risk theory T_j, and a query date t_q, PLRE predicts

$$y_{ij} = \mathbf{1}[P, p_i, \mathcal{P}_{t_q} \models T_j]. \tag{4}$$

Unlike symmetric similarity, PLRE is directional. A pair (p_i, T_j) is promoted to the risk register only when (i) the entailment score exceeds the validation-set threshold, (ii) the top supporting precedents remain valid at t_q, and (iii) the report includes mitigation text grounded in at least three cited authorities when available.

PLRE gold labels are constructed from anonymized product-attribute cards and docket-grounded theory cards, which the legal-research annotators authored from public product documentation and complaint allegations under a fixed guideline. Annotators judge a directional pair as positive only when the product attribute, legal theory, query date, and at least one valid supporting authority jointly establish exposure; negative pairs comprise ordinary mismatches and stale-authority distractors. Critically, the validity judgment encoded in the labels is produced by annotators reading docket and opinion text, independently of VIGIL's automatically extracted treatment graph, so the gate must recover label validity rather than presuppose it. Product archetypes are tracked in the split manifest, and year and litigation type are stratified to reduce temporal and topical leakage. The formulation is system-agnostic: PLRE is directional entailment subject to an exogenous, time-indexed support constraint.

Attribute	Value
Total cases	1,247
Total documents	8,934
Coverage period	2020–2025
Unique defendants	43
Risk labels	3 (High/Med/Low)
Status: Initial / Active	298 / 695
Status: Settled / Dismissed	148 / 106
Product-mapping GT cases	312

Table 2: VIGIL-Dataset summary statistics. Status counts sum to 1,247.

4 Dataset

VIGIL-Dataset is a curated benchmark of U.S. AI-related litigation designed for structured risk analysis.

4.1 Construction and Statistics

The dataset compiles CourtListener/RECAP filings and supplementary legal-news signals spanning January 2020 to December 2025, with the majority of filings concentrated in 2023–2025. Cases were obtained via a retrospective keyword search of RECAP complaints and were manually screened to confirm that AI development, training, or deployment was at issue. The scope is intentionally broader than copyright-only trackers: it includes privacy, publicity, product-liability, and contract theories when AI systems are material to the dispute. One case corresponds to one docket: related and consolidated actions are retained as separate records, so counts are dockets, not disputes. Table 2 summarizes the final annotated set (1,247 dockets after screening and deduplication); merging on dispute identity yields approximately 980 distinct disputes. The top-three defendants account for 41.3% of dockets, reflecting the concentration of high-profile generative-AI litigation; we release the defendant-frequency table so that users can assess applicability to smaller vendors.

4.2 Annotation and Availability

An automated pre-labeler first assigns a triage score using an additive keyword scoring matrix (supplementary material). Three legal research specialists then independently review a stratified sample of 400 cases under a fixed guideline. During adjudication, annotators see docket text, product attributes, candidate theory cards, and query dates, but neither system identity nor the pre-labeler's score. The remaining 847 cases retain automated pre-labels; a 5% spot-check yields 91% agreement

with at least one specialist. Inter-annotator agreement on the 400-case sample is mean pairwise Cohen's $\kappa = 0.74$ (Fleiss' $\kappa = 0.72$). Because retained labels can inflate metrics, we release per-record provenance, restrict all primary claims to the human-only evaluation, and treat retained-label results solely as scale-oriented comparisons. The dataset, guidelines, split files, seeds, and evaluation scripts are supplied as anonymized supplementary material.

5 Experiments

We evaluate the validity gate along three research questions:

- **RQ1:** Does validity gating improve Product–Litigation Risk Entailment over non-VIGIL and no-gate baselines?
- **RQ2:** How much do the interval gate and treatment weighting each contribute, and how sensitive is the gain to treatment-edge errors?
- **RQ3:** Does the calibrated stopping rule control evidence-omission risk under temporal shift?

Severity triage and blinded report ratings evaluate the surrounding system rather than the gate; they are reported in Appendix A.

5.1 Experimental Setup

Dataset split and label provenance. VIGIL-Dataset is partitioned 70/10/20 into train/validation/test splits stratified by year and litigation type (not by dispute identity; Limitations). Because many case labels come from an automated pre-labeler, case-level results use only human-adjudicated cases. Conformal calibration draws on human-adjudicated cases from the training and validation pools, kept disjoint from the test cases. The training split is used only to select few-shot exemplars and tune thresholds; no LLM is fine-tuned.

PLRE pairs (1,856 total; 28.4% positive; 19.7% stale-authority distractors) are partitioned 1,112 / 372 / 372 (train / validation / test; ≈60/20/20), stratified by product archetype and year so that every archetype appears in every split. Few-shot exemplars and the entailment threshold use only train/validation; all reported metrics use the 372 held-out test pairs.

System	P	R	F1
Product–theory cosine	0.64	0.66	0.65
LegalBERT NLI	0.69	0.68	0.68
COLIEE-style, no gate	0.71	0.70	0.70
RAG-only generation	0.73	0.74	0.73
VIGIL w/o gating	0.72	0.73	0.72
VIGIL Full	**0.77**	**0.80**	**0.78**

Table 3: PLRE comparison against external legal-NLI and precedent-retrieval baselines (LegalBERT NLI: product–theory NLI). Held-out test pairs only.

Baselines. For PLRE we compare against baselines that isolate suspected confounds: product–theory cosine, LegalBERT product–theory NLI, COLIEE-style (Rabelo et al., 2022) precedent retrieval without interval closure, RAG-only generation, and VIGIL without temporal gating. All LLM-based systems share the same GPT-4o backbone and decoding settings.

Automatic evaluation. All reported automatic metrics freeze the treatment graph produced by the extractor; low-confidence edges are not human-reviewed at evaluation time. We report PLRE precision/recall/F1, the stale-citation rate, edge-error propagation, and evidence-coverage omission risk under calibrated thresholds.

5.2 RQ1: Product–Litigation Risk Entailment

PLRE uses anonymized product-attribute cards (five archetypes) paired with docket-grounded theory cards across 312 product-mapping cases (1,856 pairs; 28.4% positive; 19.7% stale-authority distractors; $\kappa = 0.71$). Table 3: VIGIL Full reaches 0.78 F1 (+0.06 vs. no-gate; +0.05–0.13 versus non-VIGIL product–theory and legal-NLI baselines). VIGIL without gating (0.72) sits slightly below RAG-only (0.73), suggesting that the agentic loop without a validity filter can reinforce stale but lexically close precedent; in this ablation, the PLRE gain comes with the gate. Gains partly reflect rejection of stale distractors (19.7% of pairs); this version reports F1 on the full held-out set only, not on the non-distractor subset (Limitations). Primary claims rest on the 372 held-out pairs.

5.3 RQ2: Temporal Gate Ablation and Edge Errors

Table 4 decomposes the RQ1 gain into the two mechanisms of Eq. (3). Full validity-interval gating plus treatment weighting yields PLRE F1 of 0.78 and a stale-citation rate of 4.1%. Removing

Variant	F1	ΔF1	Stale %
VIGIL Full (gate + weight)	0.78	—	4.1
− treatment weighting $w(\cdot)$	0.75	−0.03	7.8
− interval gating (flat sim.)	0.72	−0.06	12.7

Table 4: Temporal precedent-graph ablation (main claim). Both mechanisms of Eq. (3) are removed in sequence; the bottom row recovers flat semantic similarity. F1 is PLRE F1; Stale is the stale-citation rate, i.e., citations to already-closed precedent ($t_q \geq e_v$).

treatment weighting alone drops F1 by 0.03 and roughly doubles stale citations (7.8%). Removing interval gating entirely (recovering flat semantic similarity) widens the gap to $\Delta\text{F1} = -0.06$ and raises the stale-citation rate to 12.7%—a three-fold increase. Without the gate, retrieval admits more lexically similar but already-closed authorities, altering the evidence that reaches the generator. This controlled degradation constitutes the paper's main methodological result.

Negative-treatment error propagation was measured by substituting gold labels on the 600 annotated spans and by additionally masking extracted edges (Appendix D). The extractor operates at the observed miss rate (≈23%, from 0.77 recall) and costs 2 PLRE F1 points relative to gold-span substitution (0.80, stale 3.7%). Masking a further 30% of extracted negative-treatment edges (≈46% effective miss rate) returns F1 to the no-gate level (0.72) with 11.3% stale citations, so the gate's benefit depends on treatment-edge quality. Stale citations persist even with gold edges because a closed authority can enter context through precedent chaining (§3.2) rather than as gated support, and because annotator validity judgments can disagree with interval closure on boundary cases.

5.4 RQ3: Evidence-Coverage Risk Control and Temporal Shift

Calibrating the evidence threshold to $\alpha \in \{0.05, 0.10, 0.20\}$ yields omission risk 0.043/0.082/0.171 in random splits. Under a temporal split (2023–24 → 2025) the achieved risk rises to 0.061/0.113/0.196, exceeding the target at $\alpha = 0.05$ and 0.10, which motivates rolling re-calibration (Appendix F).

6 Related Work

Legal NLP and domain models. Legal NLP has studied judgment prediction (Chalkidis et al., 2019), statutory reasoning (Holzenberger et al., 2020), legal benchmarks such as LexGLUE (Chalkidis et al., 2022) and LegalBench (Guha et al., 2023), case retrieval and entailment (Rabelo et al., 2022), and summarization (Huang et al., 2021; Manor and Li, 2019), often with domain-adapted encoders including LEGAL-BERT (Chalkidis et al., 2020), CaseHOLD (Zheng et al., 2021), and Lawformer (Xiao et al., 2021). Audits show that LLMs and commercial legal RAG tools still hallucinate (Dahl et al., 2024; Magesh et al., 2025). Prior work also studies overruling detection as a classification task over case text (Zheng et al., 2021); we instead extract treatment edges to build time-indexed validity intervals and apply them as a retrieval gate. These lines of work take legal text as the premise, treat the corpus as static, and do not test whether retrieved authority remained operative at a given date. PLRE changes the inference unit: product attributes are mapped to legal-risk theories in a directional, product-conditioned, and temporal manner.

Time-sensitive QA and temporal IR. Time-sensitive QA conditions answers on the moment a question is asked (Chen et al., 2021; Zhang and Choi, 2021; Liška et al., 2022; Kasai et al., 2023); temporal IR uses document time as a ranking signal (Campos et al., 2014; Dhingra et al., 2022). Temporally aware RAG adds time to retrieval by scoring or reranking evidence against document or fact timestamps (Gade et al., 2025; Qian et al., 2024). All of these treat time as a property of facts, documents, or queries. In PLRE, time enters through supersession: a later document alters the evidential status of an earlier one. Commercial citators such as Shepard's (LexisNexis, 2024) and KeyCite (Thomson Reuters, 2024) track negative treatment; citation-network analysis and edge labeling characterize the underlying graph (Fowler et al., 2007; Sadeghian et al., 2018). VIGIL attaches validity intervals to extracted treatment edges and applies them as a retrieval-time gate.

RAG, knowledge conflicts, and agents. RAG typically executes a fixed retrieval plan before generation (Lewis et al., 2020; Karpukhin et al., 2020; Gao et al., 2023); work on knowledge conflicts and outdated parametric knowledge shows that generators may prefer stale but familiar evidence unless retrieval is constrained (Longpre et al., 2021; Chen et al., 2022; Xie et al., 2024). Agent frameworks benefit from iterative tool use yet stop when the model itself deems its answer satisfactory (Yao

et al., 2023; Shinn et al., 2023; Wei et al., 2022; Park et al., 2023; Wang et al., 2024a,b). VIGIL ties stopping to an externally inspectable predicate over evidence support and mitigation coverage, and employs conformal risk control (Angelopoulos et al., 2024, 2025; Geifman and El-Yaniv, 2017) narrowly for evidence omission rather than legal correctness.

Foundation models, copyright, and fair use. Foundation-model training and output raise copyright and fair-use questions that motivate product-facing risk mapping (Henderson et al., 2023; Levendowski, 2018; Grimmelmann, 2016; Bommasani et al., 2021). Infrastructure resources such as CourtListener (Free Law Project, 2023) and the Caselaw Access Project (Harvard Law School Library, 2024) enable public-corpus construction of treatment layers that commercial citators already maintain privately.

7 Discussion

The central empirical result is that a validated treatment layer applied as a hard retrieval gate changes PLRE behavior under matched conditions: stale citations fall from 12.7% to 4.1% and F1 rises by 0.06 relative to the identical pipeline without gating (Table 4). Independent checks place negative-treatment micro-F1 at 0.80 (gold spans) and 0.81 (commercial citator). These figures are point estimates; we treat them as evidence for the mechanism rather than as significance-tested superiority, and we report no confidence intervals in this version (Limitations).

Two controls bound the claim. First, without the gate the agentic loop alone does not improve PLRE (0.72 vs. RAG-only 0.73), which suggests that extra test-time compute alone does not explain the gain (a compute-matched self-refinement control on PLRE remains to be run). Second, a surface-form keyword heuristic recovers many explicit OVERRULES/SUPERSEDES mentions but almost no material narrowing (negative-treatment F1 0.52 vs. 0.80; Appendix B), and citator-corrected intervals flip only 3.8% of held-out pairs—the extracted layer already tracks the commercial reference for downstream decisions. We deliberately do not evaluate pure date filters or learned temporal re-rankers on PLRE in the present study; a date filter cannot recover supersession endpoints from later opinions, which is the extraction problem this work targets. External-stack transfer (Protocol P1) remains a pilot that fails its pre-specified criterion (Appendix H) and is not claimed.

Severity triage and report ratings (Appendix A) evaluate the surrounding system on small samples ($n = 82$ and 15) and are secondary to PLRE.

8 Conclusion

We introduced PLRE and VIGIL, whose extracted treatment layer gates retrieval by operative authority. On held-out pairs, gating lifts PLRE F1 by 0.06 and cuts stale citations from 12.7% to 4.1% relative to flat similarity; these are point estimates whose confidence intervals, together with a non-distractor and a date-filter comparison, are the next step. The layer reaches negative-treatment micro-F1 0.80 on gold spans and 0.81 against a commercial citator.

Limitations

What this paper does and does not claim. Our primary claim is mechanistic: an extracted treatment layer, validated against gold spans and a commercial citator (negative-treatment micro-F1 0.80 / 0.81), can be applied as a hard validity gate at retrieval time. Doing so changes PLRE outcomes and stale-citation rates relative to the identical pipeline without the gate (Table 4: F1 0.72 $\rightarrow$ 0.78; stale-cite 12.7% $\rightarrow$ 4.1%). We report point estimates and do not claim statistical significance for any comparison. The contribution is the validated gate construction and its downstream effect under matched retrieval, not a leaderboard win under every possible baseline.

Statistical uncertainty. All comparisons are point estimates without confidence intervals or significance tests. The primary PLRE gap (0.78 vs. 0.72 F1 on 372 pairs) and the severity gap (0.79 vs. 0.75 on 82 cases; Appendix A) are small enough that paired bootstrap intervals are needed before either can be called reliable. Run-level mean$\pm$std over five seeds (Appendix F) vary decoding and exemplar order only and are not a substitute for per-instance uncertainty.

Baselines not evaluated on PLRE. B5 (iteration-matched self-refinement), token-matched refinement, pure document-date filters, learned temporal re-rankers, and post-hoc citation filtering (applying the same intervals after generation and regenerating) were not run on the PLRE task, so the benefit of gating before rather than after generation is not measured. A pure date filter cannot encode supersession by later opinions (the endpoint e_v is not a

property of document v), so it is not a substitute for treatment extraction; we leave a head-to-head comparison to follow-on work. B5 on severity labels is a triage control only. B1 uses additive scores only, whereas the retained pre-labels used the composite bonus, so B1 is not scored against its own labels; the full-test column of Table 5 nonetheless mixes in pre-labels and is reported for scale only. The no-gate ablation already isolates the contribution of the validity indicator under otherwise identical retrieval and generation.

Distractor composition. Stale-authority distractors constitute 19.7% of PLRE pairs by design: they test whether the gate removes closed support that lexical retrieval would otherwise promote. Gains partly reflect rejection of these distractors, and we do not report F1 on the non-distractor subset, so the effect of the gate on ranking among still-open authorities is not isolated.

Split structure. Case and PLRE splits are stratified by year and litigation type (or product archetype), not by dispute identity. Related dockets of one dispute can therefore cross the train/test boundary. Primary PLRE metrics use held-out pairs only, with few-shot exemplars and thresholds taken from train/validation, but a dispute-blocked re-split remains desirable and is listed as future work. With only five product archetypes, leave-one-archetype-out evaluation would further stress generalization and is also left to future work.

Extractor quality, transfer, and raters. Negative-treatment recall is 0.77 on gold spans (citator missed-action rate 20.4%). Protocol P1 (external-stack transfer of the gating modifier) is reported only as a pilot; the pilot does not meet the pre-specified acceptance criterion and we therefore make no transfer claim. Annotators and report raters were institutional staff; external replications can use the released guidelines. Conformal coverage weakens under temporal shift, motivating rolling recalibration in deployment. The dataset is U.S.-federal and skewed toward large defendants (top three account for 41.3% of cases).

Reproducibility. Primary experimental runs were conducted after the December 2025 cutoff of the dataset window. The treatment graph, PLRE labels, and all automatic metrics were computed on the frozen corpus that ends 31 December 2025. Controller and Analyzer calls used the OpenAI API snapshot `gpt-4o-2024-08-06`. Hosted snapshots can be retired, so bit-identical re-runs require that snapshot; the Llama-3.1-70B + bge run shows the pipeline also runs on open weights. Because many underlying opinions pre-date the knowledge cutoff of both models, we cannot rule out that either had seen parts of the corpus during pre-training; the open-weights run does not remove this confound.

Future work. Larger human adjudication, paired-bootstrap CIs on the main tables, non-distractor and dispute-blocked audits, B5/token-matched, date-filter, post-hoc-filter and temporal re-ranker controls on PLRE, leave-one-archetype-out evaluation, and broader jurisdictional coverage.

Ethical Considerations

Decision-support boundary. VIGIL emits risk intelligence, evidence summaries, and mitigation candidates with provenance and residual uncertainty for qualified human reviewers. It is not legal advice: it does not replace counsel, predict litigation outcomes, or recommend business decisions, and severity labels are triage proxies that are not interpreted as liability. Every High- or Medium-risk item is presented for review by a qualified person, and unresolved legal-status atoms are surfaced as such rather than resolved automatically.

Risks of misuse and over-reliance. A risk register that looks authoritative invites over-reliance: a missed negative-treatment edge can let stale authority support a report, and a Low label can be read as clearance. Because the corpus is U.S.-federal and dominated by a few large defendants, labels may under-represent theories relevant to smaller vendors or other jurisdictions. We therefore surface provenance and unresolved legal-status atoms with every item and do not position severity labels as legal conclusions.

Human oversight of failure modes. Low-confidence negative-treatment edges enter a human review queue, since missed negative-treatment actions are the failure mode most likely to admit stale authority into a report. Reviewers can distinguish a genuine product–theory match from a retrieval gap or an unresolved legal-status ambiguity.

Data governance. Released data are public docket metadata and machine-generated summaries derived from CourtListener/RECAP; we

do not redistribute attorney-authored complaint text beyond what is already public, and the P2 citator study releases only the cross-walk, sampling seeds, and agreement statistics, with no redistributed proprietary citator content. The released dataset is distributed under a CC-BY-4.0 license for the machine-generated annotations and metadata; underlying docket text remains subject to the original CourtListener/RECAP terms of use. Released records contain the names of parties and counsel as they appear in these public filings; we collect no other personal information, and news items are represented only by links and derived metadata. The release is intended for research on legal-risk analysis, not for automated legal decisions. Protocol P2 used a commercial citator under the authors' institutional research subscription. Because the precise product name is omitted for licensing reasons, exact reproduction of the P2 numbers requires an equivalent institutional subscription. We release the mapping rules so that any subscriber can re-run the agreement calculation.

Annotators and raters. The three legal research specialists who adjudicated labels and the two specialists who rated reports are staff of the authors' institution and performed this work within their regular paid duties. Before annotation began, participants received a written description of the task, the intended research use of their labels, data-handling practices, and the voluntary nature of participation; verbal consent was obtained and recorded. The work consisted solely of labeling public court documents and collected no personal information about the participants beyond institutional employment status. Under the authors' institutional human-subjects policy for secondary analysis of public records with no identifiable private information about research subjects, the activity was determined not to constitute human-subjects research requiring IRB review. This determination is recorded in the released annotation guidelines. Institutional affiliation of annotators is a potential source of bias (Limitations). Annotation guidelines are released with the dataset so that external replications can use independent raters.

Use of AI assistants. AI writing assistants were used to edit and polish the text of this paper. An AI coding assistant also wrote the code for the P1 pilot, following a design fixed by the authors; the pilot design was frozen before any run, and its results were reported as produced. Research ideas, system design, and the remaining experiments and analyses are the authors' own.

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A Severity Triage and Report Quality

These experiments evaluate the surrounding risk-analysis system rather than the validity gate, and their samples are small; we report them for completeness.

Baselines. B1 Additive-only Rule (the pre-labeler's scoring matrix without its composite bonus, thresholds tuned on validation); B2 Single-Pass LLM; B3 RAG-only; B4 Single-Agent (VIGIL without the goal-conditioned loop); and

B5 Iteration-Matched Self-Refinement (Reflexion-style (Shinn et al., 2023), matched on VIGIL's mean budget of 2.4 iterations, not on tokens). B2–B5 share the same GPT-4o backbone and decoding settings as VIGIL. Severity-label F1 measures triage-label agreement over three labels (High/Medium/Low), not legal liability.

Severity agreement. Table 5 reports macro severity-label agreement on the human-adjudicated test cases ($n = 82$). VIGIL reaches 0.79 macro F1 vs. 0.75 for iteration-matched B5; the same rank order holds on the full held-out test set, which includes retained pre-labels and is reported for scale only. With $n = 82$ we do not treat this four-point gap as established, and it is not significance-tested.

System	Human-adj.			Full
	F1	P	R	F1
B1 Additive-only	0.55	0.55	0.56	0.62
B2 Single-Pass LLM	0.70	0.70	0.71	0.72
B3 RAG-only	0.72	0.72	0.72	0.74
B4 Single-Agent	0.74	0.74	0.74	0.76
B5 Self-Refinement	0.75	0.75	0.75	0.77
VIGIL Full	**0.79**	**0.79**	**0.80**	**0.81**

Table 5: Macro severity-label agreement. Human-adjudicated test cases ($n = 82$). Full: held-out test including retained pre-labels.

Report quality. Two blinded specialists rated VIGIL and B2–B5 reports on 15 cases (75 reports) along Accuracy, Completeness, Actionability, Evidence Sufficiency, and Clarity (5-point scales). VIGIL's mean scores are 4.3/4.1/4.2/4.4/3.9 on the five scales, highest on Evidence Sufficiency. Mean inter-rater $\kappa = 0.68$ (0.62–0.74 across scales). With $n = 15$ we make no pairwise superiority claim over B2–B5; any Wilcoxon signed-rank comparison on per-case means is exploratory and uncorrected for multiplicity.

Backbone dependence. Replacing GPT-4o with an open-weights Llama-3.1-70B (Grattafiori et al., 2024) + bge (Xiao et al., 2024) stack, unchanged otherwise, lowers full-test-set severity macro F1 from 0.81 to 0.76 (mean over five runs) with 2.6 vs. 2.4 mean iterations (Table 13). Baselines were not re-run on this stack, so this shows that VIGIL runs without the proprietary model, not that its advantage is backbone-independent.

B Treatment-Edge Extraction

Table 6 reports per-relation accuracy of the treatment-edge extractor (§3.4) against the 600 hand-labeled gold citation spans. Negative-treatment edges drive interval closure (SUPERSEDES/OVERRULES) or down-weighting (material NARROWS) and are micro-averaged separately from the high-frequency CITES relation.

Relation	P	R	F1
CITES	0.94	0.96	0.95
DISTINGUISHES	0.81	0.76	0.78
NARROWS (material)	0.79	0.71	0.75
SUPERSEDES	0.85	0.78	0.81
OVERRULES	0.88	0.82	0.85
Negative treatment (micro)	0.84	0.77	0.80

Table 6: Treatment-edge extraction accuracy on the 600 gold citation spans.

Keyword heuristic baseline. A surface-form heuristic flags citation sentences containing *overruled*, *superseded*, *abrogated*, *overturned*, or *no longer good law*. On the same 600 spans its negative-treatment micro P/R/F1 is 0.71/0.41/0.52 (LLM classifier: 0.84/0.77/0.80). It recovers many explicit OVERRULES/SUPERSEDES mentions but almost no material NARROWS, and cannot label DISTINGUISHES polarity.

C Citator Agreement (Protocol P2)

Design. P2 compares the extracted negative-treatment edges with a commercial citator. A stratified random sample of edges from the graph (1,812 nodes, 4,760 edges), oversampling negative treatment, was fixed by a released seed before any citator lookup. Citator signals were mapped to VIGIL's relation taxonomy through a pre-specified crosswalk by two annotators blind to VIGIL's predictions, with a third adjudicating. Two acceptance thresholds were frozen before sampling: negative-treatment micro-F1 ≥ 0.80, and a missed-action rate no higher than the extractor's internal-gold miss rate (≈23%). No citator content is redistributed; the crosswalk, seeds, and agreement statistics are released.

Results. The audit sampled 400 treatment candidates, 162 of them negative treatment. Crosswalk agreement between annotators was $\kappa = 0.78$; on the negative-vs.-other dichotomy, VIGIL–citator

κ is 0.74. Table 7 gives edge agreement and Table 8 action agreement and downstream impact; both thresholds are met. Of 156 predicted negative-treatment actions, 129 match the citator relation; of the 27 relation-level false positives, only 7 change the closing or down-weighting action. Material NARROWS remains the weakest relation and stays in the human-review queue.

Relation	n	P	R	F1
OVERRULES	54	0.87	0.83	0.85
SUPERSEDES	47	0.82	0.79	0.80
material NARROWS	61	0.80	0.77	0.78
Neg. treatment (micro)	162	0.83	0.80	0.81
Cautionary	78	0.81	0.74	0.77
Positive/neutral	160	0.93	0.95	0.94

Table 7: P2 edge agreement with the citator-derived reference; n is the number of reference positives.

Metric	Value
Matched negative-treatment actions	129 / 162
Missed-action rate	20.4%
False-action rate (7 / 156)	4.5%
Action-date error: median / mean	18 / 31 days
Action-date error: 90th percentile	74 days
Point-in-time validity accuracy	0.88
PLRE decision flips after correction	3.8%

Table 8: P2 action agreement (closure for OVERRULES/SUPERSEDES, down-weighting for material NARROWS). Decision flips are a share of all held-out PLRE pairs.

D Edge-Error Propagation

Table 9 measures how negative-treatment errors propagate to PLRE. The gold row substitutes gold labels on the 600 annotated spans only; the other extracted edges are unchanged. Mask rows additionally remove a random fraction of the extracted negative-treatment edges before retrieval.

Setting	Eff. miss	F1	ΔF1	Stale %
Gold edges (on spans)	0%	0.80	+0.02	3.7
Extractor output	≈23%	0.78	—	4.1
+ mask 10%	≈31%	0.76	−0.02	6.2
+ mask 23%	≈41%	0.74	−0.04	8.9
+ mask 30%	≈46%	0.72	−0.06	11.3

Table 9: Edge-error propagation (PLRE F1, held-out pairs). Effective miss rate $\approx 1 - 0.77(1 - x)$ for mask fraction x; ΔF1 is relative to the extractor row.

E Agentic-Loop and Retriever Ablations

Table 10 removes one loop component at a time (full held-out test set, including retained pre-labels; mean over five runs). Only removing query expansion lowers retrieval recall. Table 11 compares retrievers on the 200-pair development set used to choose the default (§3.2).

Variant	Severity F1	Recall@10
VIGIL (Full)	0.81	0.84
− Goal Checker	0.76	0.84
− Planner (fixed order)	0.78	0.84
− query expansion	0.79	0.76

Table 10: Agentic-loop component ablation. Removing the Goal Checker leaves a single agent; removing the Planner fixes the task order.

Retriever	Recall@10
Dense (`text-embedding-3-large`)	0.79
LegalBERT (fine-tuned on RECAP)	0.81
Hybrid (BM25 + neural re-ranking)	0.84

Table 11: Retriever comparison on the development set.

F Stopping Rule, Calibration, and Backbones

Proof of Observation 1 (first clause). Let $\Phi(R_t) \in \mathbb{Z}_{\geq 0}$ count unmet atomic sub-conditions of $\mathcal{G}$: one per missing severity label, per under-supported evidence requirement, and per missing mitigation. (A1) *Monotone evidence caching*: query expansion unions new hits with the prior evidence cache, so a satisfied evidence or mitigation atom stays satisfied and $\Phi(R_{t+1}) \leq \Phi(R_t)$. (A2) *Attainable progress*: if $\Phi(R_t) > 0$ and some unmet atom is attainable (its evidence exists in $\mathcal{D}$), the Planner schedules it and it is resolved, so $\Phi(R_{t+1}) \leq \Phi(R_t) - 1$. Hence, if every atom is attainable, Φ reaches 0 within $\Phi(R_0)$ steps; otherwise the loop stops at $T_{\max}$ and the Goal Checker returns the residual atoms for human escalation. □ (A1) holds by construction; (A2) is a design assumption that we do not verify directly.

Conformal calibration (second clause). For calibration cases $c_1, \ldots, c_n$ ($n = 150$), let $L_i(\tau) \in \{0, 1\}$ indicate that a retrievable, human-labeled High-risk theory of c_i is omitted, and let $\hat{R}_n(\tau)$ be its mean, which does not increase as τ decreases. Following conformal risk control (Angelopoulos

et al., 2024), $\hat{\tau} = \sup\{\tau : \frac{n}{n+1}\hat{R}_n(\tau) + \frac{1}{n+1} \le \alpha\}$ gives $\mathbb{E}[L_{n+1}(\hat{\tau})] \le \alpha$ for an exchangeable test case. For joint (k, τ) selection we use a Learn-then-Test grid (Angelopoulos et al., 2025) with fixed-sequence testing, which yields $\Pr[R(\hat{k}, \hat{\tau}) \le \alpha] \ge 1 - \delta$ instead. Table 12 gives achieved risk: random splits meet every budget, while the temporal split exceeds it at $\alpha \in \{0.05, 0.10\}$.

Split	α=0.05	α=0.10	α=0.20
Random	0.043	0.082	0.171
Temporal	0.061	0.113	0.196

Table 12: Achieved evidence-omission risk under random resplits and a temporal split (2023–24 → 2025); evidence coverage, not severity correctness.

Backbones. Table 13 reports the full system under two backbones; baselines were not re-run on the open-weights stack, so the table supports no cross-backbone comparison with them.

Configuration	Macro F1	Mean iters
GPT-4o stack	0.81 ± 0.02	2.4 ± 0.3
Llama-3.1-70B stack	0.76 ± 0.02	2.6 ± 0.3

Table 13: Full VIGIL under two backbones: GPT-4o with `text-embedding-3-large`, and Llama-3.1-70B with `bge-large` (severity macro F1 on the full held-out test set; mean ± std over five runs).

G Hyperparameters and Evaluation Protocol

Runs and seeds. The five seeds vary controller/Analyzer decoding (temperature 0.2) and few-shot exemplar order; the retrieval index, treatment graph, edge classifier (temperature 0), and test membership are fixed. Run-level means are not per-instance uncertainty estimates. The Goal Checker is deterministic rule evaluation over R_t (no LLM call).

Report ratings. Two specialists, blind to system identity, rated 75 reports (the same 15 cases under each of five systems, presented in random order with identifiers stripped). Inter-rater κ is 0.68/0.71/0.65/0.74/0.62 on Accuracy, Completeness, Actionability, Evidence Sufficiency, and Clarity.

Parameter	Value	Description
k	3	Min. documents per risk item
τ	0.62	Default min. relevance (re-ranker score scaled to $[0, 1]$); recalibrated in RQ3
$T_{\max}$	5	Max. loop iterations
Controller, Analyzer	GPT-4o	Temperature 0.2
Edge classifier	GPT-4o	Temperature 0
Embedding dim.	3072	`text-embedding-3-large`
Mapping threshold	0.72	Product–risk bi-encoder similarity
w: positive / open	1.00	Positive or no negative treatment
w: cautionary	0.75	Cautionary or questioned
w: material narrows	0.50	Materially narrowed
w: closed	0.00	Overruled or superseded; removed by Eq. (3)

Table 14: Loop and gating hyperparameters used in all reported experiments.

H External-Transfer Pilot (Protocol P1)

Protocol. P1 asks whether the gate of Eq. (3), applied as post-hoc re-scoring, improves external retrievers that use none of VIGIL's loop, data, or goal predicate. Its criterion, frozen before any run, requires Holm-significant nDCG@10 gains and stale@10 reductions on at least two of three hosts, with non-inferiority (margin -0.5 nDCG points) on control queries. The full protocol has not been executed; we report a pilot.

Pilot. Gold closures come from the Constitution Annotated table of overruled U.S. Supreme Court decisions (Congressional Research Service, 2022), matched to Caselaw Access Project opinions (Harvard Law School Library, 2024); overrulings decided in or after 2020 are excluded, so the pilot is disjoint from the VIGIL-Dataset window. This leaves 253 pairs and 29,030 opinions. Queries are citing passages from later opinions with case names, citations, and years masked: post-closure queries cite the overruling case (984 queries, 208 pairs), pre-closure queries cite the later-overruled case before its overruling (1,061), and control queries cite cases absent from the table (1,133). Hosts are BM25 (Robertson and Zaragoza, 2009), `bge-small-en-v1.5` (Xiao et al., 2024), and LegalBERT (Chalkidis et al., 2020), each retrieving the top 100 opinions decided before the query date. The pilot deviates from P1: it has no COLIEE-style

slice and no extractor-treatment variant, and it uses a cluster bootstrap by gold pair.

Result. The pilot does not support transfer: only bge meets the criterion (Table 15). With gold intervals the gate removes every overruled case from the top 10 on all hosts, non-inferiority holds everywhere, and on pre-closure queries the gate never removed the gold case. nDCG@10 gains are at most 0.13 points, because removal helps only when the stale case outranks the correct one. LegalBERT is a weak host here and rarely retrieves the stale case. On partial overrulings, $w \in \{0.25, 0.5, 0.75\}$ gives identical rankings, so the pilot is uninformative about $w(\cdot)$. A post-hoc exclusion of four BM25 queries whose gold case was itself later overruled would let two hosts pass; because it was chosen after seeing results, the pre-specified verdict stands. The queries, relevance judgments, gold closures, corpus identifiers, and a scorer are released as a supersession-aware retrieval testbed.

Host	nDCG@10	Stale@10	ΔnDCG@10	ΔStale@10 (pts)	Ctrl LB	Met
BM25	28.88	8.3%	$+0.09$ $[-0.58, +0.52]$	-8.3^{*} $[-10.7, -6.2]$	$+0.04$	no
bge-small	8.35	5.5%	$+0.13^{*}$ $[+0.04, +0.27]$	-5.5^{*} $[-7.1, -3.9]$	$+0.00$	yes
LegalBERT	2.30	0.4%	$+0.00$ $[+0.00, +0.00]$	-0.4^{*} $[-0.8, -0.1]$	$+0.00$	no

Table 15: P1 pilot, gold-treatment variant, on post-closure queries of full overrulings ($n = 984$, 208 pair clusters). nDCG@10 and Stale@10 are for the ungated host; Δ is gated minus ungated with 95% cluster-bootstrap CIs (10,000 resamples); *Holm-adjusted $p < 0.05$ across hosts. Ctrl LB: lower bound of ΔnDCG@10 on control queries (non-inferiority margin -0.5).